Mouse database v2: what data is available and what is in it

`data_exploration.ipynb` (v1) describes the database but analyses only trail 3. This notebook looks at **all trails** and at **both exports**: the 2023 export (`/data/mice/maus.pandas.pkl`, converted from `maus_mysql.sql`) and the database dump of 2025-08-08 (`mausprojekt-3.sql`), converted with `tfmplayground/mouse/sql_dump.py`.

- **Part A** compares the two exports and decides which one is the base.
- **Part B** describes all trails of the new export.
- **Part C** checks how usable each trail is for weight forecasting.
- **Part D** looks at the new tables (lab samples, the application log) and at columns v1 never used.
- **Part E** summarises data quality and lists candidate datasets for modelling.

No models are trained here. Every number in the text cells is computed in the notebook.

0. Setup

The worktree that holds this notebook comes first on `sys.path`, so `tfmplayground` (including `sql_dump.py`) is imported from here and not from the editable install of the main checkout.

```
tfmplayground imported from /workspaces/nanoPFN/src/TFM-Playground/.claude/worktree
s/bridge-cse_01Kxsthj9LqeaqKaNs4ZbzvC/tfmplayground/__init__.py
```

English tables

All tables, columns, categorical values and free text are read from the English copy of the exports (`mausprojekt-3.en.pkl`; the 2023 export is translated in memory with the same layer, `sql_dump.translate`). The translation fails on any table, column, value or free-text entry it does not know. The German-schema pickles are only handed to the existing loaders (`michael.load_dataset` and its step-by-step replay in C1), which need the original schema.

Glossary for matching the database owners' vocabulary: English table name and the original table name.

table	original name	table	original name
anaesthesia	Narkosen	mri_weightings	MRTWichtungen
app_destinations	Destinationen	organ_systems	Organsysteme
app_log	Logbuch	organs	Organe
app_modules	Module	paraffin_samples	Paraffin
app_settings	SystemVariablen	pcr	PCR
blood_samples	Blut	pcr_import	PCR_Import
body_sides	SeitenBez	pressure_measurements	Druckmessungen
bone_marrow_samples	Knochenmark	procedures	Prozeduren
breeders	Zuechter	projects	Tierversuche
cell_lines	Tumorzelllinien	qualifications	Qualifikationen
chemo_drugs	ChemoMed	radiotherapy	Bestrahlungen
chemo_protocols	ChTKonzepte	roles	Rollen
chemotherapy	Chemotherapie	routes	Applikation
cohorts	Kohorten	rt_protocols	RTKonzepte
colours	Farben	sectioning_procedures	OTSchnittProzeduren
cryo_samples	Kryo	sectioning_protocols	OTSchnittKonzept
daily_device_checks	Tagescheck	sites	Topographie
death_causes	Todesursachen	slide_box_types	OTBoxTypen
drug_vehicles	Traeger	slide_boxes	OTBoxen
injection_batches	VGruppe	slides	Objekttraeger
institutions	KlinikFirma	stainings	Faerbungen
ionisation_chambers	Messkammern	strains	Tiertypen
kill_methods	Toetungsarten	surgeries	Operationen
locations	Standorte	tissue_samples	Gewebe
marking_types	Markierungen	trails	Versuchsreihen
mice	Versuchstiere	tumours	Tumor
mri_contrast_agents	MRTKontrast	visits	Visiten
mri_sequences	MRTSequenzen		

The free-text columns (one remark per record, surgery reports, the request log of the web application) were machine-translated: every unique original string was translated once by an AI model (Claude), without a translation library, and the translations are kept next to the data in `mausprojekt-3.free_text_en.json` (not in the repository, as the remarks may name staff). Numbers, units, doses, sample codes, names and initials are kept as written; fragments the model could not read are kept in the original as `[?de: ...]` (count below). In the request log, parameter names and values are translated and credentials masked. The originals are in the German-schema pickle `mausprojekt-3.pandas.pkl`. The free text is not analysed in this notebook.

22 free-text columns: 1,944 unique strings per column (1,286 distinct overall; for the request log the unique text values of its parameters), 1 of them kept partly in the original as uncertain.

table	column	rows with text	translated strings	content
radiotherapy	remark	220	24	remarks on single RT fractions
blood_samples	remark	42	12	remarks on blood samples
chemotherapy	remark	153	5	remarks on single ChT administrations
tissue_samples	remark	6770	219	remarks on tissue samples
bone_marrow_samples	remark	3	3	remarks on bone-marrow samples
cryo_samples	remark	2830	171	remarks on cryo samples
app_log	payload	54509	492	raw request parameters of the web application (credentials masked)
mri_sequences	remark	2	2	remarks on MRI sequences
anaesthesia	remark	83	19	remarks on anaesthesia sessions
surgeries	report	67	13	surgery reports
pcr	remark	689	9	remarks on PCR results
paraffin_samples	remark	3639	196	remarks on paraffin samples
procedures	finding	13	12	clinical findings
procedures	procedure	13	11	clinical procedures performed
app_settings	value	14	13	configuration values of the web application
daily_device_checks	remark	4	4	remarks on device checks
tumours	remark	973	20	remarks on tumour injections
injection_batches	remark	75	75	description of each injection batch
trails	remark	1	1	remark on one trail
mice	remark	344	191	remarks on single mice
visits	burden_note	14	9	burden notes on single visits
visits	remark	4171	443	remarks on single visits

Plot style and helpers (from v1)

Tidy frames

v1's setup cells, wrapped in `build_frames(tables)` so that they run on both exports: `mice`, `visits`, `radio`, `chemo`, `surgery`, `anaesthesia`, `tumours`, plus the lookups, all from the English tables.

Load both exports

The 2023 export is translated in memory. Its conversion named two long-text columns `longtext` and kept their line breaks escaped (see `sql_dump.py`); they get their real names first.

```
old: 25 tables, 843 mice, 22142 visits, 15 trails
new: 55 tables, 988 mice, 28125 visits, 18 trails, last visit 2025-06-24
```

Part A: the two exports side by side

A1. Table inventory

Rows and columns of every table in both exports.

table	category	status	rows old	rows new	added rows	cols old	cols new	content
visits	core	both	22142	28125	5983	41	41	one row per visit: body weight, relative weight, burden score sheet, tumour volume
mice	core	both	843	988	145	29	29	one row per mouse: trail, role, sex, strain, birth and death date, cause of death, protocols, batch
anaesthesia	treatment	both	2823	3114	291	10	10	anaesthesia sessions
radiotherapy	treatment	both	1958	2158	200	18	18	one row per radiotherapy fraction
chemotherapy	treatment	both	1316	1406	90	13	13	one row per chemotherapy administration
surgeries	treatment	both	47	67	20	9	9	surgeries (removal of the primary tumour)
procedures	treatment	both	13	13	0	7	7	clinical procedures (fluids, soft food, ...)
tumours	outcome	both	844	973	129	9	9	tumours per mouse: site, cell line, injection and removal date, weight
injection_batches	design	both	66	75	9	6	6	injection batches: cohort, cell line, date
cohorts	design	both	27	31	4	6	6	cohorts: trail x cell line x strain
trails	design	both	15	18	3	15	15	trails: name, aims, start/end, project
projects	design	both	12	15	3	11	11	approved projects with licence number
cell_lines	lookup	both	19	21	2	6	6	lookup table
rt_protocols	lookup	both	20	20	0	10	10	lookup table
chemo_protocols	lookup	both	18	19	1	15	15	lookup table
sites	lookup	both	15	15	0	4	4	lookup table
roles	lookup	both	11	14	3	3	3	lookup table
strains	lookup	both	9	10	1	7	7	lookup table
death_causes	lookup	both	8	8	0	3	3	lookup table
chemo_drugs	lookup	both	6	7	1	7	7	lookup table
routes	lookup	both	6	6	0	3	3	lookup table
kill_methods	lookup	both	5	5	0	3	3	lookup table
body_sides	lookup	both	4	4	0	2	2	lookup table
breeders	lookup	both	4	4	0	6	6	lookup table
daily_device_checks	admin	both	213	232	19	14	14	daily quality check of the irradiation device
tissue_samples	lab/sample	only new		7362			10	tissue taken at necropsy: site, side, date, weight
paraffin_samples	lab/sample	only new		4500			12	paraffin-embedded samples: sample type and storage position
cryo_samples	lab/sample	only new		3733			15	cryo-preserved samples: sample type and storage position
pcr	lab/sample	only new		3134			20	ALU-PCR results: cell counts and Cp values per sample
bone_marrow_samples	lab/sample	only new		1034			9	bone-marrow samples per mouse: date, running number
blood_samples	lab/sample	only new		1028			9	blood samples per mouse: date, running number
mri_sequences	lab/sample	only new		2			8	MRI sequences per mouse
pressure_measurements	lab/sample	only new		2			6	pressure measurements (mmHg) per mouse and site
slides	lab/sample	only new		1			9	microscope slides: staining, metastasis counts
pcr_import	lab/sample	only new		0			18	staging table for PCR imports (empty)
app_log	logging	only new		176922			6	log of the web application: logins, page views and submitted forms
organs	lookup	only new		33			4	lookup table
institutions	lookup	only new		15			10	lookup table
organ_systems	lookup	only new		12			3	lookup table
marking_types	lookup	only new		11			5	lookup table
colours	lookup	only new		9			3	lookup table
stainings	lookup	only new		8			4	lookup table
qualifications	lookup	only new		7			3	lookup table
mri_weightings	lookup	only new		5			3	lookup table
drug_vehicles	lookup	only new		4			3	lookup table

table	category	status	rows old	rows new	added rows	cols old	cols new	content
ionisation_chambers	lookup	only new		3			7	lookup table
locations	lookup	only new		2			5	lookup table
mri_contrast_agents	lookup	only new		2			4	lookup table
slide_box_types	lookup	only new		2			6	lookup table
sectioning_procedures	lookup	only new		1			6	lookup table
sectioning_protocols	lookup	only new		1			4	lookup table
slide_boxes	lookup	only new		1			7	lookup table
app_modules	admin	only new		61			10	menu entries of the web application
app_settings	admin	only new		14			3	configuration of the web application
app_destinations	admin	only new		2			3	redirect targets of the web application

```
25 shared tables, 30 only in the new export, 0 only in the old one
Shared tables with different columns: none
visits columns in the new dump: 40 | visits.timestamp in the new dump: False | in t
he old dump: False
```

`visits.timestamp` is in neither dump: the 2023 conversion derived it from the visit date and time, and `sql_dump.py` derives it the same way (its self-check compares it with the 2023 export). Neither loader reads it:

```
visits.timestamp used by michael.py or preprocessing.py: False
```

A2. Is the old export a subset of the new one?

Every shared table is joined on its primary key (the first column, checked to be unique in both). A row is *unchanged* if all shared columns are equal (missing = missing), *changed* otherwise, *deleted* if its key is only in the old export and *added* if only in the new one.

table	key	rows old	rows new	unchanged	changed	deleted	added	changed columns
anaesthesia	anaesthesia_id	2823	3114	2823	0	0	291	
body_sides	side_id	4	4	4	0	0	0	
breeders	breeder_id	4	4	4	0	0	0	
cell_lines	cell_line_id	19	21	19	0	0	2	
chemo_drugs	drug_id	6	7	6	0	0	1	
chemo_protocols	chemo_protocol_id	18	19	18	0	0	1	
chemotherapy	administration_id	1316	1406	1312	4	0	90	route_id (4)
cohorts	cohort_id	27	31	27	0	0	4	
daily_device_checks	check_id	213	232	213	0	0	19	
death_causes	death_cause_id	8	8	8	0	0	0	
injection_batches	batch_id	66	75	62	4	0	9	cells (4)
kill_methods	kill_method_id	5	5	4	1	0	0	name (1)
mice	mouse_id	843	988	803	40	0	145	death_date (38), death_cause_id (38), kill_method_id (38), active (2)
procedures	procedure_id	13	13	13	0	0	0	
projects	project_id	12	15	12	0	0	3	
radiotherapy	fraction_id	1958	2158	1958	0	0	200	
roles	role_id	11	14	11	0	0	3	
routes	route_id	6	6	6	0	0	0	
rt_protocols	rt_protocol_id	20	20	20	0	0	0	
sites	site_id	15	15	15	0	0	0	
strains	strain_id	9	10	9	0	0	1	
surgeries	surgery_id	47	67	47	0	0	20	
trails	trail_id	15	18	13	2	0	3	name (2)
tumours	tumour_id	844	973	844	0	0	129	
visits	visit_id	22142	28125	22142	0	0	5983	

The changed rows in detail. In `mice`, the change per mouse and trail:

change	trail 0	1	15	Total
death date, cause and killing method added	0	0	38	38
other: active	1	1	0	2
Total	1	1	38	40

table	key	column	old	new
chemotherapy	1286	route_id	1.000	2
chemotherapy	1300	route_id	1.000	2
chemotherapy	1311	route_id	1.000	2
chemotherapy	1312	route_id	1.000	2
injection_batches	59	cells	PC-3 RGB + luciferase	PC-3 RGB + luciferase 1
injection_batches	60	cells	PC-3 RGB + luciferase	PC-3 RGB + luciferase 2
injection_batches	61	cells	PC-3 RGB + luciferase	PC-3 RGB + luciferase 3
injection_batches	62	cells	PC-3 RGB + luciferase	PC-3 RGB + luciferase 4
kill_methods	1	name	CO2	CO2/O2
trails	11	name	Pilot PC3	Pilot N127/2019 PC3
trails	12	name	Main experiment PC3	Main experiment N127/2019 PC3

```
Changed values in free-text columns: 0
```

Does the loader see the same trail-3 data in both exports? `michael.load_dataset` with v1's configuration (regression check against v1: 186 mice, 3,049 rows).

```
trail 3, old export: 186 mice, 3049 rows
trail 3, new export: 186 mice, 3049 rows
The trail-3 study cohort is identical in both exports.
all trails, roles 1-3, old export: 434 mice, 12397 rows
```

A3. Which export is the base?

What this tells us (Part A).

- The new export has 55 tables instead of 25: all 25 old tables with the same columns, plus 30 new ones. `visits.timestamp` is a column the 2023 conversion added, not one the database lost; the loaders do not use it.
- **The old export is contained in the new one:** 0 rows deleted, 0 of 22,142 visits changed, 5,983 visits and 145 mice added. The edits to old rows are completions and relabelling: 38 mice of trail(s) 15 got their death date, cause and killing method (this changes what the loader keeps for trail 15), 2 mice changed their `active` flag, and 11 values in other tables changed (chemotherapy.route_id 4, injection_batches.cells 4, kill_methods.name 1, trails.name 2), i.e. chemotherapy routes from "unknown" to i.p. and renamed trails, batches and killing method.
- The trail-3 study cohort through `michael.load_dataset` is identical in both exports (186 mice, 3,049 rows), so v1 and the results stay valid on the new export.
- **Base for the rest of the notebook: the new export.**

Part B: all trails (new export)

B1. One row per trail

Split in three tables to fit the page: design, population, then visits, treatments and outcomes.

trail	name	licence	start	end	project
0	(not in the trails table)				
1	Pilot 1	G14/13	2013-06-17		Influence of radiotherapy, chemotherapy, simultaneous radiochemotherapy and surgery on metastasis in xenograft mouse models
2	Cell migration under irradiation	G133/12	2013-07-24		Selectin-dependent interaction of cell populations in a clinically relevant human tumour model in the mouse under therapy
3	Main experiment 1	G14/13	2013-08-03		Influence of radiotherapy, chemotherapy, simultaneous radiochemotherapy and surgery on metastasis in xenograft mouse models
4	Pilot Loges 1	G36/13	2013-11-27		Resistance and bone-marrow micrometastasis under anti-androgen therapy of breast cancer
5	Pilot 1	G			Pilot with 2 animals - learning methods
6	Breast cancer model		2014-06-10		
7	Nanoparticles	G133/12	2015-08-27		Selectin-dependent interaction of cell populations in a clinically relevant human tumour model in the mouse under therapy
8	Pilot 1 cell migration	G14/13	2018-01-08		Influence of radiotherapy, chemotherapy, simultaneous radiochemotherapy and surgery on metastasis in xenograft mouse models
9	Pilot 1 olaparib	G67/16	2018-05-31		Radiosensitisation of PTEN- or CHD1-depleted prostate carcinomas by PARP1 inhibition
10	Main experiment TAS102	N013/2018	2018-05-01		TAS102
11	Pilot N127/2019 PC3	N127/2019	2021-04-13	2022-04-13	Influence of locally ablative radiotherapy, chemotherapy, radiochemotherapy and surgery on metastasis in two xenograft mouse prostate models
12	Main experiment N127/2019 PC3	N127/2019	2021-04-16		Influence of locally ablative radiotherapy, chemotherapy, radiochemotherapy and surgery on metastasis in two xenograft mouse prostate models
13	Pilot N006/2020	N006/2020	2021-06-22		Immunomodulation by radiotherapy in glioblastoma
14	Main experiment N006/2020	N006/2020	2021-06-23	2021-06-23	Immunomodulation by radiotherapy in glioblastoma
15	Main experiment N052/2020	N052/2020	2021-09-17	2021-10-17	Relevance of clonal heterogeneity for the cerebral tropism and radiotherapy response of metastatic and primary brain tumours
16	Main experiment N029/2023	N029/2023	2023-08-08		Tracking of systemic drugs, nanoparticles and immune cells in a xenograft mouse prostate tumour model with X-ray fluorescence imaging
17	Pilot N068/2024	N068/2024	2024-11-22		Functional role of small-RNA clusters in hepatocarcinogenesis
18	Organ removal for ALU-PCR	ORG1148/2025	2025-04-23		Organ-removal application supplementing project "Tracking of systemic drugs, nanoparticles and immune cells in a xenograft mouse prostate tumour model with X-ray fluorescence imaging" (N029/2023)

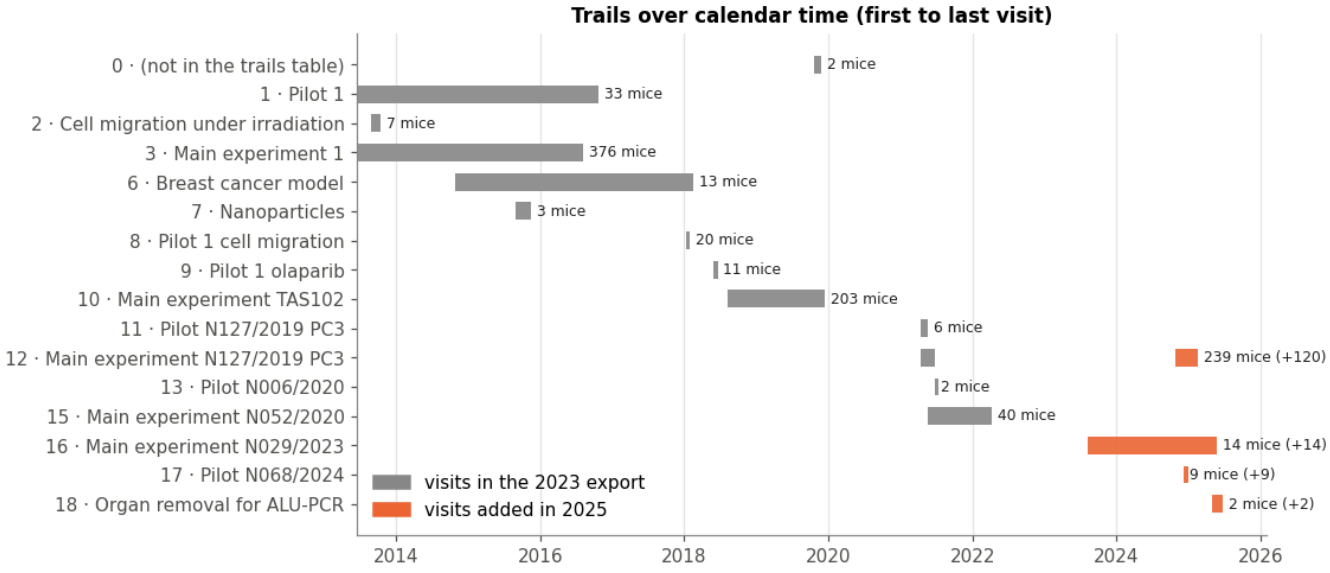
trail	mice	mice per role	male / female	strain	cell line
0	2	RT 1, – 1	1 / 1	NSG NOD.Cg-Prkdcscid, PFP/RAG2	SW48
1	33	ChT 21, RChT 9, Obs 2, RT 1	25 / 8	PFP/RAG2, SCID	H69 RGB, OH-1 RGB, PC-3 RGB
2	7	RT 7	7 / 0	SCID, SCID Select	H69AR RGB, OH-1 RGB
3	376	C-end 73, RT 73, RChT 61, C-start 54, ChT 54, NoTumour 38, Surgery 18, Obs 3, – 2	237 / 139	PFP/RAG2, SCID	H69 RGB, LNCaP RGB, OH-1 RGB
4	6	RT 6	0 / 6	Balb/c	
5	2	Pilot 2	0 / 2	PFP/RAG2	HT29
6	13	RT 12, RChT 1	1 / 12	FVB/N-Tg(MMTV-PyVT), SCID	PymT
7	3	RT 3	0 / 3	PFP/RAG2	OH1
8	20	C-start 6, C-end 6, RT 5, Obs 3	20 / 0	SCID	H69AR RGB
9	11	RT 4, RChT 4, ChT 2, C-end 1	11 / 0	NSG NOD.Cg-Prkdcscid	C5
10	203	RChT 76, ChT 48, RT 40, C-end 32, – 5, NoTumour 2	91 / 112	PFP/RAG2	CACO-2, HCT116, HT29, SW48
11	6	RT 4, ChT 1, RChT 1	6 / 0	SCID	PC-3 RGB + luciferase
12	239	Surgery 44, RChT 40, C-start 40, RT 39, C-end 39, ChT 36, NoTumour 1	239 / 0	SCID	PC-3 RGB + luciferase, PC-3DR RGB + luciferase
13	2	RT 2	2 / 0	C57Bl/6	GL261
14	0				
15	40	RT 18, C-end 13, NoTumour 7, ChT 1, C-start 1	1 / 39	NMRI-Foxn1nu, SCID	NCH644OBC, PC-3 RGB + luciferase
16	14	ChT 10, Macro 3, Nano 1	14 / 0	SCID	PC-3 RGB + luciferase
17	9	Pilot 9	9 / 0	C57Bl/6J	Plasmid induction CTNNB1 P53 MYC-LUC
18	2	OrganRem 2	2 / 0	SCID	PC-3 RGB + luciferase

trail	first visit	last visit	visits	visits per mouse	RT fractions	ChT admin.	surgeries	% with death date	causes of death
0	2019-10-21	2019-11-25	35	35.000	3	0	0	100.000	Experiment ended as planned 1, Unknown 1
1	2013-06-17	2016-10-24	1217	25.000	36	84	0	94.000	Killed: poor general condition 12, Spontaneous death 8, Poor general condition / suspected lymphoma 4, Experiment ended as planned 3, Unknown 3, Killed: tumour ulceration 1
2	2013-08-28	2013-10-16	7	1.000	7	0	0	0.000	
3	2013-06-17	2016-08-08	7047	15.000	666	192	19	100.000	Experiment ended as planned 296, Killed: poor general condition 40, Unknown 17, Spontaneous death 11, Unknown: disposed of by animal care 5, Poor general condition / suspected lymphoma 4, Death directly after surgery 2, Killed: tumour ulceration 1
4			0		9	0	0	0.000	
5			0		0	0	0	100.000	Experiment ended as planned 1, Killed: tumour ulceration 1
6	2014-10-27	2018-02-15	13	1.000	12	0	0	100.000	Experiment ended as planned 12, Unknown 1
7	2015-08-27	2015-11-16	39	13.000	15	0	0	100.000	Experiment ended as planned 3
8	2018-01-08	2018-01-29	428	23.000	15	0	0	100.000	Experiment ended as planned 20
9	2018-05-31	2018-06-14	11	1.000	8	6	0	100.000	Experiment ended as planned 11
10	2018-08-06	2019-12-16	7154	33.000	816	958	0	96.000	Experiment ended as planned 169, Killed: tumour ulceration 25, Unknown 1
11	2021-04-15	2021-05-21	192	30.000	13	3	0	100.000	Experiment ended as planned 6
12	2021-04-15	2025-02-16	10062	41.000	393	151	48	100.000	Experiment ended as planned 225, Spontaneous death 6, Killed: poor general condition 5, Death directly after surgery 3
13	2021-06-22	2021-06-22	2	1.000	2	0	0	0.000	
14			0		0	0	0		
15	2021-05-20	2022-04-07	1286	16.500	163	2	0	100.000	Experiment ended as planned 37, Spontaneous death 3
16	2023-08-08	2025-05-25	501	33.000	0	10	0	100.000	Experiment ended as planned 14
17	2024-12-10	2024-12-10	9	1.000	0	0	0	0.000	
18	2025-04-29	2025-06-24	116	58.000	0	0	0	100.000	Experiment ended as planned 2

B2. What is new since the 2023 export

Mice and visits per trail in both exports, and the trails over calendar time: grey is the time span of the visits already in the old export, orange that of the added visits.

trail	mice old	mice new	visits old	visits new	mice added	visits added	new trail
12	119	239	4705	10062	120	5357	False
16	0	14	0	501	14	501	True
17	0	9	0	9	9	9	True
18	0	2	0	116	2	116	True



B3. Protocols per trail

Radiotherapy protocols (from the fractions in `radiotherapy`) and chemotherapy drugs and protocols (from `chemotherapy`), with the number of mice. The schedule is fractions x dose per fraction of the protocol (the protocols also differ in beam energy, 200 vs 225 kV). The last columns show how much of each trail uses the same RT schedule and ChT drugs as trail 3.

trail	RT protocols (mice)	RT schedules (mice)	RT fractions per mouse	RT total dose (Gy)	% RT mice with trail-3 schedule
0	d1-5 2.0 Gy 1	5 x 2 Gy 1	3 [3–3]	9 [9–9]	0.000
1	d1-5 10.0 Gy 8	5 x 10 Gy 8	5 [5–5]	50 [48–50]	100.000
2	d1 10.0 Gy 6, d1 15.0 Gy 1	1 x 10 Gy 6, 1 x 15 Gy 1	1 [1–1]	10 [10–10]	0.000
3	d1-5 10.0 Gy 134	5 x 10 Gy 134	5 [5–5]	50 [50–50]	100.000
4	d1 6 Gy - femora 3, d1 6 Gy - mammary gland 3	2 x 6 Gy 3, 1 x 6 Gy 3	2 [1–2]	9 [6–12]	0.000
6	Mammary02 d1 10.0 Gy 11, Mammary01 d1 3.0 Gy 1	1 x 10 Gy 11, 1 x 3 Gy 1	1 [1–1]	10 [10–10]	0.000
7	d1-5 10.0 Gy 3	5 x 10 Gy 3	5 [5–5]	50 [50–50]	100.000
8	d1-4 10.0 Gy 1, d1 10.0 Gy 1, d1-3 10.0 Gy 1, d1-5 10.0 Gy 1, d1-2 10.0 Gy 1	4 x 10 Gy 1, 1 x 10 Gy 1, 3 x 10 Gy 1, 5 x 10 Gy 1, 2 x 10 Gy 1	3 [2–4]	30 [20–40]	20.000
9	d1 2.0 Gy 8	1 x 2 Gy 8	1 [1–1]	2 [2–2]	0.000
10	d1-10 3.0 Gy 68, d1-5 2.0 Gy 40, d1 2.0 Gy 4, d1 3.0 Gy 4	10 x 3 Gy 68, 5 x 2 Gy 40, 1 x 2 Gy 4, 1 x 3 Gy 4	10 [3–10]	20 [9–30]	0.000
11	d1-5 10.0 Gy 225 kV 3, d1-5 10.0 Gy 2	5 x 10 Gy 5	1 [1–5]	10 [10–50]	100.000
12	d1-5 10.0 Gy 225 kV 69, d1-5 10.0 Gy 10	5 x 10 Gy 79	5 [5–5]	50 [50–50]	100.000
13	d1 10 Gy225 kV 2	1 x 10 Gy 2	1 [1–1]	10 [10–10]	0.000
15	d1-10 2.0 Gy 10, d1-6 6x2.0 Gy 8	10 x 2 Gy 10, 6 x 2 Gy 8	10 [6–12]	19 [12–24]	0.000
16					

trail	ChT drugs (mice)	ChT protocols (mice)	ChT admin. per mouse	ChT routes	% ChT mice with trail-3 drug
1	Cisplatin 19, Etoposide 6, Docetaxel 3	BC standard (old) 9, Doctaxel 20 mg/kg d1-8-15 3, Cisplatin 6mg/kg d1 2, Cisplatin dose 1/3 2, Cisplatin dose 2/3 2, Etoposide dose 3/3 2, Etoposide dose 1/3 2, Etoposide dose 2/3 2, Cisplatin + pause + etoposide 2, Cisplatin reduced + pause + etoposide 2	3 [3–4]	i.p.	79.000
3	Cisplatin 76, Docetaxel 40	Cisplatin 6mg/kg d1 76, Doctaxel 20 mg/kg d1-8-15 40	1 [1–3]	i.p.	100.000
9	Olaparib 6	Olaparib 2 h before RT 6	1 [1–1]	oral gavage	0.000
10	TAS102 70, Capecitabine 53	TAS102 d1-10 52, Capecitabine d1-10 27, Capecitabine d1-5 26, TAS102 d1-5 18	10 [5–10]	oral gavage, unknown	0.000
11	Docetaxel 2	Docetaxel d1,8 4 mg/kg 1, Docetaxel D1 8mg/kg 1	2 [1–2]	i.p.	100.000
12	Docetaxel 76	Docetaxel d1,8 4 mg/kg 76	2 [2–2]	i.p.	100.000
15	Docetaxel 1	Docetaxel d1,8 4 mg/kg 1	2 [2–2]	i.p.	100.000
16	Ruthenium glutamine 10	Ruthenium glutamine d1 10	1 [1–1]	i.p.	0.000

rt_protocol_id	name	Gy per fraction	total Gy	fractions	per day	per week	beam	remark
1	d1-5 10.0 Gy	10	50	5	1	5	200 kV	Standard 5 x 10 Gy
2	d1 10.0 Gy	10	10	1	1	1	200 kV	Standard 1 x 10 Gy
3	d1 15.0 Gy	15	15	1	1	1	200 kV	Collaboration cell migration
4	d1 6 Gy - mammary gland	6	6	1	1	1	200 kV	Collaboration breast irradiation
5	d1 6 Gy - femora	6	12	2	2	2	200 kV	Collaboration breast irradiation
6	Mammary01 d1 3.0 Gy	3	3	1	1	1	200 kV	Collaboration breast cancer
7	Mammary02 d1 10.0 Gy	10	10	1	1	1	200 kV	Collaboration breast cancer
8	d1-4 10.0 Gy	10	40	4	1	4	200 kV	Standard 4 x 10 Gy
9	d1-3 10.0 Gy	10	30	3	1	3	200 kV	Standard 3 x 10 Gy
10	d1-2 10.0 Gy	10	20	2	1	2	200 kV	Standard 2 x 10 Gy
11	d1 2.0 Gy	2	2	1	1	1	200 kV	Pilot olaparib
12	d1-10 3.0 Gy	3	30	10	1	5	200 kV	Standard 10 x 3 Gy
13	d1-5 2.0 Gy	2	10	5	1	5	200 kV	Standard 5 x 2 Gy
14	d1 3.0 Gy	3	3	1	1	1	200 kV	Standard 1 x 3 Gy
15	d1-10 2.0 Gy	2	20	10	1	5	200 kV	Standard 10 x 2 Gy
17	d1-5 10.0 Gy 225 kV	10	50	5	1	5	225 kV	Standard 5 x 10 Gy
18	d1 10 Gy225 kV	10	10	1	1	1	225 kV	Standard 1 x 10 Gy 225 kV
19	d1-11 12x2.0 Gy	2	24	12	2	6	225 kV	Special case 10 x 2.0 Gy + 2 x 2.0 Gy
20	d1-6 6x2.0 Gy	2	12	6	1	6	225 kV	Standard 6 x 2 Gy

chemo_protocol_id	protocol	drug 1	dose 1 (mg/kg)	days 1	drug 2	dose 2 (mg/kg)	days 2	remark
1	BC standard (old)	Cisplatin	6	1	Etoposide	12.000	3	
2	Doctaxel 20 mg/kg d1-8-15	Docetaxel	20	3				
3	Cisplatin dose 1/3	Cisplatin	2	1				
4	Cisplatin dose 2/3	Cisplatin	4	1				
5	Cisplatin 6mg/kg d1	Cisplatin	6	1		0.000		
6	Etoposide dose 1/3	Etoposide	4	3				
7	Etoposide dose 2/3	Etoposide	8	3				
8	Etoposide dose 3/3	Etoposide	12	3				
9	Cisplatin + pause + etoposide	Cisplatin	6	1	Etoposide	12.000	3	Cisplatin d1, etoposide d3-5
10	Cisplatin reduced + pause + etoposide	Cisplatin	4	1	Etoposide	12.000	3	Cisplatin d1, etoposide d3-5
12	Olaparib 2 h before RT	Olaparib	1	1				
13	TAS102 d1-10	TAS102	150	10				TAS102 150 mg/kg body weight d1-10
14	Capecitabine d1-5	Capecitabine	400	5				Capecitabine d1-5
15	TAS102 d1-5	TAS102	150	5				TAS102 150 mg/kg body weight d1-5
16	Capecitabine d1-10	Capecitabine	400	10				Capecitabine 400 mg/kg body weight d1-10
17	Docetaxel d1,8 4 mg/kg	Docetaxel	4	2		0.000		Docetaxel d1,8 4mg/kg
18	Docetaxel D1 8mg/kg	Docetaxel	8	1				
19	Ruthenium glutamine d1	Ruthenium glutamine	200	1	Ruthenium glutamine			Ruthenium glutamine d1

B4. Roles 12–14 (new in 2025)

role	meaning
12	OrganRem: organ removal
13	Macro: injection of macrophages
14	Nano: injection of nanoparticles

role trail	Macro	Nano	OrganRem	Total
16	3	1	0	4
18	0	0	2	2
Total	3	1	2	6

Rows per table for the mice of these roles:

role_abbr	mice	visits	radiotherapy	chemotherapy	surgeries	anaesthesia	tumours	tissue	blood	bone marrow	cryo	paraffin	pc
Macro	3	97	0	0	0	0	3	0	0	0	0	0	
Nano	1	32	0	0	0	0	1	0	0	0	0	0	
OrganRem	2	116	0	0	0	0	0	16	2	2	6	8	

id	trail	role	visits	first	last	weight
985	16	13	32	2025-04-29	2025-05-24	18.100
986	16	13	31	2025-04-29	2025-05-24	18.900
987	16	14	32	2025-04-29	2025-05-25	17.900
988	16	13	34	2025-04-29	2025-05-25	20.000
989	18	12	58	2025-04-29	2025-06-24	20.050
990	18	12	58	2025-04-29	2025-06-24	23.150

What this tells us (Part B).

- The new export has 18 trails (3 new: 16, 17, 18). Most of the growth is trail 12, which went from 119 to 239 mice and from 4,705 to 10,062 visits; the new trails are small (trail 16: 14 mice, trail 17: 9 mice, trail 18: 2 mice).
- Trail 3 remains the largest trail (376 mice); the only other trails with more than 100 mice are 10 (203), 12 (239).
- Trails in which at least half of the irradiated mice got trail 3's RT schedule (5 x 10 Gy): 1, 3, 7, 11, 12. Trails in which at least half of the ChT mice got a trail-3 drug (Cisplatin, Docetaxel): 1, 3, 11, 12, 15. The other trails ran different protocols (e.g. fractionated 2–3 Gy schedules, TAS102/capecitabine) and are different experiments.
- Roles 12–14 cover 6 mice in trail(s) 16, 18 with 245 visits (0.9% of all visits); their weights were recorded but they got no RT/ChT, and the organ-removal mice have necropsy samples only.

Part C: usability per trail for weight forecasting

C1. Cohort flow per trail

Replays `michael.load_dataset` step by step (as v1 C1), for all trails at once, on the original-schema export the loader reads. Each column is the number of mice left after that filter. "lost to trail start" counts mice that would have had enough weights if the records before the trail start or birth were not dropped. The final column is checked against `michael.load_dataset`.

Roles RT/RChT/ChT — kept: 519 mice, 16,376 rows (old export: 434 mice, 12,397 rows). *treated*: RT, ChT or surgery recorded.

trail	all mice	in trails table	role 1–3	death date known	treated	≥ 5 weights	weights > 0 g	weights ≤ 100 g	lost to trail start
0	2	0	0	0	0	0	0	0	0
1	33	33	31	29	29	29	26	26	0
2	7	7	7	0	0	0	0	0	0
3	376	376	188	188	188	188	187	186	0
4	6	6	6	0	0	0	0	0	0
5	2	2	0	0	0	0	0	0	0
6	13	13	13	13	13	0	0	0	0
7	3	3	3	3	3	3	3	3	0
8	20	20	5	5	5	5	5	5	0
9	11	11	10	10	10	0	0	0	0
10	203	203	164	158	156	156	155	153	0
11	6	6	6	6	6	6	6	6	0
12	239	239	115	115	115	115	115	114	0
13	2	2	2	0	0	0	0	0	0
15	40	40	19	19	18	18	18	17	1
16	14	14	10	10	10	9	9	9	0
17	9	9	0	0	0	0	0	0	0
18	2	2	0	0	0	0	0	0	0

Mice kept per trail and role:

trail	ChT	RChT	RT	Total
1	18	7	1	26
3	53	60	73	186
7	0	0	3	3
8	0	0	5	5
10	44	71	38	153
11	1	1	4	6
12	36	39	39	114
15	0	0	17	17
16	9	0	0	9
Total	161	178	180	519

C2. Widening the role filter

The same flow with all roles. Controls and other untreated mice are then removed by the *treated* filter (Michael's loader drops mice without RT/ChT/OP), so a second run switches that filter off (`michael.UNTREATED` patched to an empty list, inside this notebook only) to see what the untreated mice could add as context.

All roles: 740 mice kept with the treatment filter, 888 mice and 26,974 rows without it.

trail	all mice	in trails table	role any	death date known	treated	≥ 5 weights	weights > 0 g	weights ≤ 100 g	lost to trail start
0	2	0	0	0	0	0	0	0	0
1	33	33	33	31	31	31	28	28	0
2	7	7	7	0	0	0	0	0	0
3	376	376	376	376	310	290	289	288	0
4	6	6	6	0	0	0	0	0	0
5	2	2	2	2	0	0	0	0	0
6	13	13	13	13	13	0	0	0	0
7	3	3	3	3	3	3	3	3	0
8	20	20	20	20	20	20	20	20	0
9	11	11	11	11	11	0	0	0	0
10	203	203	203	195	162	162	161	159	0
11	6	6	6	6	6	6	6	6	0
12	239	239	239	239	199	199	199	198	0
13	2	2	2	0	0	0	0	0	0
15	40	40	40	40	28	28	28	27	1
16	14	14	14	14	10	9	9	9	0
17	9	9	9	0	0	0	0	0	0
18	2	2	2	2	2	2	2	2	0

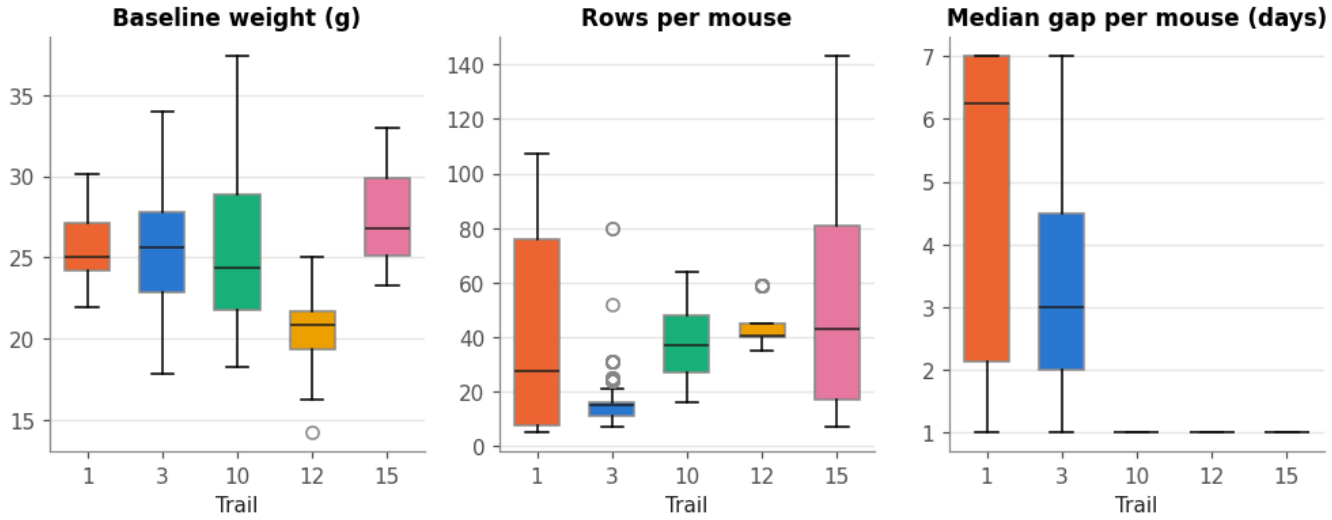
role	mice	rows	treated	length	median_gap_days	baseline_g	trails
0 –	5	150	0	30 [30–30]	1 [1–1]	31.2 [30.0–32.3]	10 5
1 RT	180	5761	180	31 [16–41]	1 [1–2]	23.5 [21.4–26.3]	3 73, 12 39, 10 38, 15 17, 8 5, 11 4, 7 3, 1 1
2 RChT	178	5812	178	34 [16–43]	1 [1–2]	23.6 [21.1–27.2]	10 71, 3 60, 12 39, 1 7, 11 1
3 ChT	163	4838	161	27 [16–40]	1 [1–3]	24.2 [21.1–27.1]	3 53, 10 46, 12 36, 1 18, 16 9, 11 1
4 C-start	86	1914	0	18 [8–32]	1 [1–6]	22.8 [20.9–25.4]	3 40, 12 40, 8 6
5 C-end	160	3908	2	22 [14–36]	1 [1–3]	23.6 [21.5–27.1]	3 73, 12 39, 10 29, 15 13, 8 6
7 Obs	6	208	0	22 [22–45]	2 [1–6]	25.8 [24.4–26.6]	8 3, 1 2, 3 1
8 NoTumour	42	1869	0	46 [13–74]	1 [1–7]	25.5 [22.7–28.7]	3 33, 15 7, 10 1, 12 1
9 Surgery	62	2294	62	37 [31–42]	1 [1–1]	21.2 [20.3–23.2]	12 44, 3 18
12 OrganRem	2	114	0	57 [57–57]	1 [1–1]	11.5 [11.3–11.7]	18 2
13 Macro	3	79	0	26 [26–26]	1 [1–1]	12.5 [11.9–13.1]	16 3
14 Nano	1	27	0	27 [27–27]	1 [1–1]	15.0 [15.0–15.0]	16 1

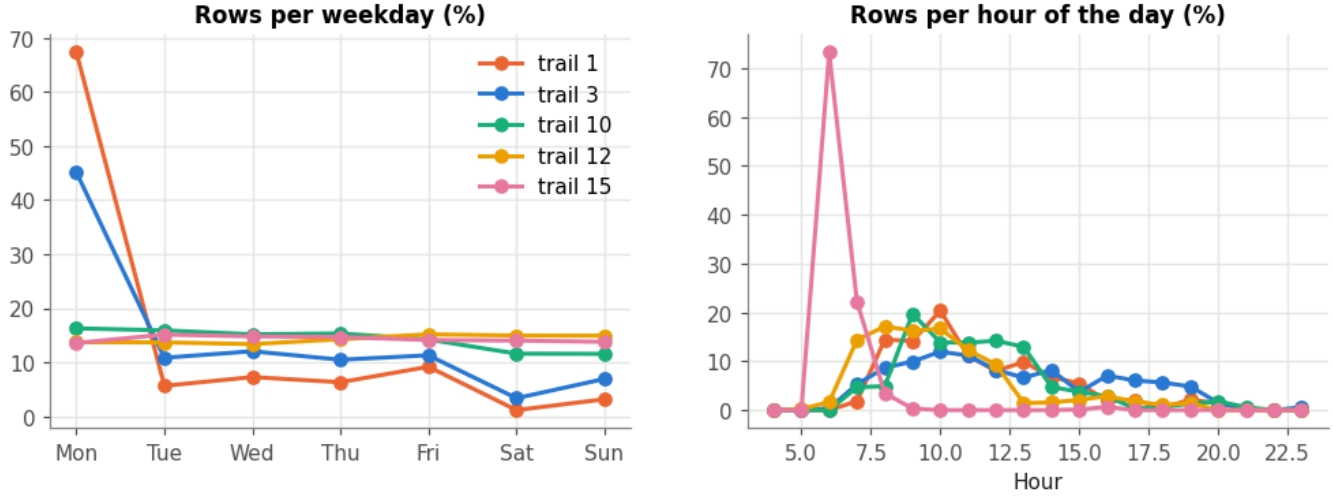
C3. Series characteristics per trail

For the roles RT/RChT/ChT through `michael.load_dataset` (`D13`), all trails. Gaps are days between consecutive rows of a mouse; loss is the lowest weight relative to the first one.

trail	mice	rows	series length	median gap (days)	% gaps 1 day	% gaps 7 days	% gaps > 7 days	% rows on Mondays
1	26	1047	28 [8–76]	7.000	15	56	4	67
3	186	3049	15 [11–16]	2.000	44	24	1	45
7	3	39	13 [13–13]	4.500	31	0	23	15
8	5	110	22 [22–22]	1.000	95	0	0	18
10	153	5808	37 [27–48]	1.000	93	0	0	16
11	6	181	29 [29–33]	1.000	95	0	0	14
12	114	4900	40 [40–45]	1.000	98	0	0	14
15	17	900	43 [17–81]	1.000	98	0	0	14
16	9	342	47 [5–62]	1.000	97	0	0	14

trail	time of day (h)	% rows without time	baseline weight (g)	step change SD (g)	% mice ≥ 10% loss	% mice ≥ 20% loss	% rows with treatment	weights before 1st treatment
1	10.9 [9.6–13.3]	0	25.1 [24.2–27.1]	0.900	54	35	8	0 [0–5]
3	12.2 [10.1–16.1]	0	25.6 [22.8–27.8]	1.020	57	22	26	7 [3–8]
7	9.6 [7.5–15.5]	0	21.3 [21.2–21.8]	0.700	0	0	38	3 [3–3]
8	9.5 [8.4–10.4]	0	24.1 [23.7–25.4]	0.470	40	0	14	14 [14–14]
10	11.7 [9.8–13.2]	0	24.4 [21.8–28.9]	0.430	21	3	20	14 [12–14]
11	8.8 [8.2–10.4]	0	21.1 [20.2–23.1]	0.620	50	17	9	22 [18–27]
12	10.0 [8.5–11.6]	0	20.8 [19.3–21.7]	0.520	13	3	10	32 [31–36]
15	6.9 [6.8–7.0]	0	26.8 [25.1–29.9]	0.730	41	12	16	0 [0–1]
16	9.6 [8.1–11.2]	0	23.0 [20.9–24.3]	0.360	0	0	3	46 [4–61]

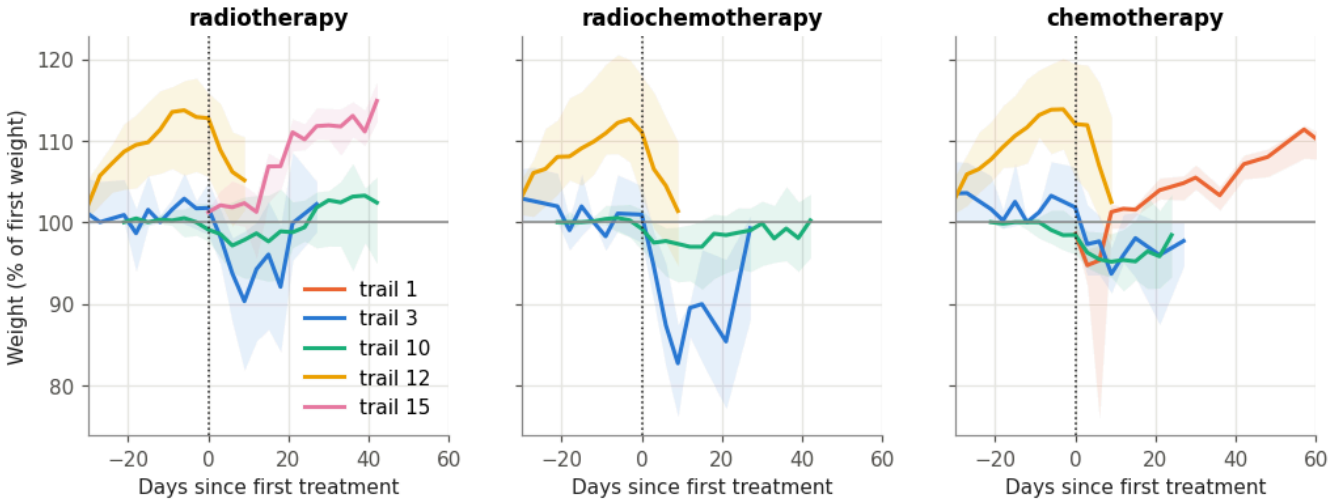




C4. Shift against trail 3

Population, era and protocol of the trails with at least `MIN_MICE` kept mice, and the weight course aligned at the first treatment (median over mice in 3-day bins, as v1 C4).

trail	mice	strain	% male	cell lines	years	roles	RT protocols	ChT drugs	days 1 row → treatm
1	26	SCID 23, PFP/RAG2 3	81	PC-3 RGB 3, OH-1 RGB 3, H69 RGB 2	2013–2013	ChT 18, RChT 7, RT 1	d1-5 10.0 Gy 8	Cisplatin 16, Etoposide 6, Docetaxel 3	0 [0–21
3	186	SCID 128, PFP/RAG2 58	66	OH-1 RGB 81, LNCaP RGB 58, H69 RGB 47	2013–2016	RT 73, RChT 60, ChT 53	d1-5 10.0 Gy 133	Cisplatin 74, Docetaxel 40	28 [21-49]
10	153	PFP/RAG2 153	43	SW48 55, HT29 49, HCT116 49	2018–2019	RChT 71, ChT 44, RT 38	d1-10 3.0 Gy 64, d1-5 2.0 Gy 37, d1 2.0 Gy 4, d1 3.0 Gy 4	TAS102 67, Capecitabine 48	14 [14-16]
12	114	SCID 114	100	PC-3DR RGB + luciferase 60, PC-3 RGB + luciferase 54	2021–2024	RChT 39, RT 39, ChT 36	d1-5 10.0 Gy 225 kV 69, d1-5 10.0 Gy 9	Docetaxel 75	32 [31-36]
15	17	NMRI-Foxn1nu 17	0	NCH644OBC 17	2021–2022	RT 17	d1-10 2.0 Gy 10, d1-6 6x2.0 Gy 7		0 [0–1]



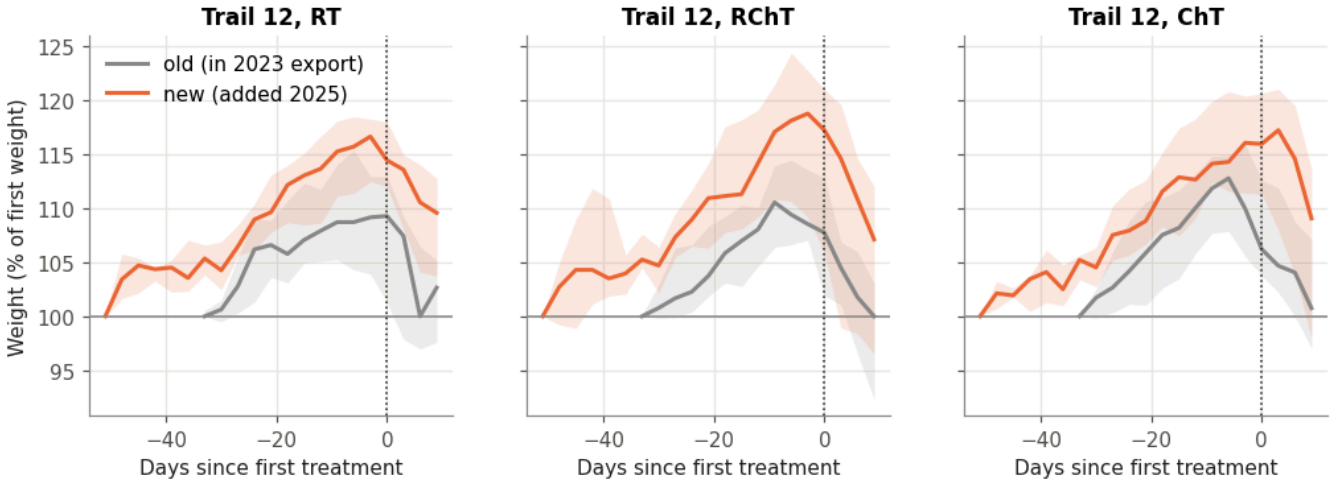
Time on study (tumour injection to death) per trail, RT/RChT/ChT mice of the kept series (v1's Kaplan–Meier plot, by trail instead of role). As in v1, almost all deaths are planned, so the curves show the follow-up design.



C5. Trail 12: old half vs new half

Trail-12 mice that are already in the 2023 export against those added since.

half trail 12	new (added 2025)	old (in 2023 export)
mice	120	119
first_visit	2024-10-24	2021-04-15
last_visit	2025-02-16	2021-06-25
cohorts	PC-3DR RGB + luciferase main experiment 120	PC-3 RGB + luciferase main experiment 119
cell_lines	PC-3DR RGB + luciferase 120	PC-3 RGB + luciferase 119
strains	SCID 120	SCID 119
roles	C-start 20, ChT 20, RChT 20, Surgery 20, RT 20, C-end 19, NoTumour 1	Surgery 24, C-start 20, C-end 20, RChT 20, RT 19, ChT 16
male	120	119
RT protocols	d1-5 10.0 Gy 225 kV 40	d1-5 10.0 Gy 225 kV 29, d1-5 10.0 Gy 10
ChT drugs	Docetaxel 40	Docetaxel 36
kept series (RT/RChT/ChT)	60	54
series length	42 [40–48]	40 [40–41]
baseline weight (g)	20.9 [19.3–22.0]	20.8 [19.6–21.6]
lowest weight (% of first)	100.0 [95.8–100.0]	96.9 [93.7–100.0]
median gap (days)	1.0	1.0



C6. Naive last-value error per trail

The error of "next weight = last weight" on the rolling one-step-ahead origins (v1 C7), a yardstick for how hard each trail is. *Overall* is the mean of the three role MAEs, as in the results tables. The rolling machinery is run once on trail 3 and must match v1 (0.695 g); per trail the same numbers are computed directly (the last-value forecast of origin k is weight $k - 1$).

Trail 3 via rolling_predictions: overall MAE 0.695 g

trail	mice	origins	MAE RT	MAE RChT	MAE ChT	MAE overall (mean of roles)	MAE all origins
1	26	995	0.996	0.536	0.561	0.697	0.564
3	186	2677	0.541	0.844	0.701	0.695	0.662
7	3	33	0.412				0.412
8	5	100	0.363				0.363
10	153	5502	0.147	0.172	0.175	0.165	0.166
11	6	169	0.450	0.484	0.500	0.478	0.466
12	114	4672	0.343	0.372	0.376	0.363	0.363
15	17	866	0.509				0.509
16	9	324			0.276		0.276

What this tells us (Part C).

- With v1's filters the new export yields 519 mice / 16,376 rows for RT/RChT/ChT over all trails (old export: 434 / 12,397); trails with at least 10 mice: 1 (26), 3 (186), 10 (153), 12 (114), 15 (17). Largest losses: role 1–3 (407 mice), ≥ 5 weights (24 mice), death date known (23 mice).
- Widening the roles with Michael's treatment filter kept gives 740 mice (+221), because controls have no treatment. Without that filter 888 mice / 26,974 rows have series with ≥ 5 weights (369 more, 246 of them controls C-start, C-end), usable as untreated context.
- The trails are **different populations**: of the other trails with ≥ 10 mice, 3 share a mouse strain and 1 share a cell line with trail 3; cadence, series length and treatment timing differ too (tables above). Pooling is possible technically but mixes experiments.
- Trail 12's added mice use cell line(s) PC-3DR RGB + luciferase against PC-3 RGB + luciferase in the old half (a different tumour model, the same RT schedule and ChT drug, visits 2021-04–2021-06 vs 2024-10–2025-02). Baseline weight 20.8 [19.6–21.6] vs 20.9 [19.3–22.0] g, lowest weight 96.9 [93.7–100.0] vs 100.0 [95.8–100.0] % of the first weight (old vs new half). It is a second experiment under one trail id and should carry its own label.
- Naive last-value MAE: trail 3 0.662 g over all origins (0.695 g as the mean of the role MAEs, as in v1); other trails range from 0.166 to 0.564 g.

Part D: new tables and unused columns

D1. The 30 tables that are new in 2025

For every new table: purpose, rows, whether it refers to mice (`mouse_id`), how many mice and trails it covers and the range of its first date column.

table	group	purpose	rows	cols	mice	% of mice
tissue_samples	lab/sample	tissue taken at necropsy: site, side, date, weight	7362	10	813	82.300
paraffin_samples	lab/sample	paraffin-embedded samples: sample type and storage position	4500	12	813	82.300
cryo_samples	lab/sample	cryo-preserved samples: sample type and storage position	3733	15	806	81.600
pcr	lab/sample	ALU-PCR results: cell counts and Cp values per sample	3134	20	641	64.900
bone_marrow_samples	lab/sample	bone-marrow samples per mouse: date, running number	1034	9	802	81.200
blood_samples	lab/sample	blood samples per mouse: date, running number	1028	9	802	81.200
mri_sequences	lab/sample	MRI sequences per mouse	2	8	2	0.200
pressure_measurements	lab/sample	pressure measurements (mmHg) per mouse and site	2	6	1	0.100
slides	lab/sample	microscope slides: staining, metastasis counts	1	9	1	0.100
pcr_import	lab/sample	staging table for PCR imports (empty)	0	18	0	0.000
app_log	logging	log of the web application: logins, page views and submitted forms	176922	6		
organs	lookup	lookup table	33	4		
institutions	lookup	lookup table	15	10		
organ_systems	lookup	lookup table	12	3		
marking_types	lookup	lookup table	11	5		
colours	lookup	lookup table	9	3		
stainings	lookup	lookup table	8	4		
qualifications	lookup	lookup table	7	3		
mri_weightings	lookup	lookup table	5	3		
drug_vehicles	lookup	lookup table	4	3		
ionisation_chambers	lookup	lookup table	3	7		
locations	lookup	lookup table	2	5		
mri_contrast_agents	lookup	lookup table	2	4		
slide_box_types	lookup	lookup table	2	6		
sectioning_procedures	lookup	lookup table	1	6		
sectioning_protocols	lookup	lookup table	1	4		
slide_boxes	lookup	lookup table	1	7		
app_modules	admin	menu entries of the web application	61	10		
app_settings	admin	configuration of the web application	14	3		
app_destinations	admin	redirect targets of the web application	2	3		

table	trails	dates	% cells missing
tissue_samples	0, 1, 3, 5, 7, 8, 9, 10, 11, 12, 15, 16, 18	2013-02 – 2025-06	8.700
paraffin_samples	0, 1, 3, 5, 7, 8, 9, 10, 11, 12, 15, 16, 18		20.900
cryo_samples	0, 1, 3, 5, 7, 8, 9, 10, 11, 12, 15, 16, 18		13.500
pcr	0, 1, 3, 8, 10, 11, 12, 15	2014-01 – 2021-10	7.800
bone_marrow_samples	0, 1, 3, 7, 8, 9, 10, 11, 12, 15, 16, 18	2013-07 – 2025-06	13.200
blood_samples	0, 1, 3, 7, 8, 9, 10, 11, 12, 15, 18	2013-07 – 2025-06	13.200
mri_sequences	1, 3	2013-07 – 2013-09	0.000
pressure_measurements	3	2013-10 – 2013-10	0.000
slides	3		33.300
pcr_import			

D2. Lab and sample tables

Coverage per trail (mice with at least one row), when the samples were taken relative to death, and the missingness per column.

trail	mice in trail	tissue	blood	bone marrow	cryo	paraffin	pcr
0	2	1	2	2	1	1	1
1	33	11	9	10	10	11	1
3	376	338	334	335	338	338	325
5	2	2	0	0	2	2	0
7	3	3	3	3	3	3	0
8	20	20	20	20	20	20	20
9	11	11	11	11	11	11	0
10	203	181	186	181	181	181	178
11	6	6	6	6	6	6	6
12	239	235	228	229	229	235	109
15	40	1	1	1	1	1	1
16	14	2	0	2	2	2	0
18	2	2	2	2	2	2	0

table	date column	rows dated	% within 1 day of death	% > 1 day before death	% > 1 day after death	median days after death
tissue_samples	sampling_date	7362	99	1	0	-0.000
blood_samples	sampling_date	1028	100	0	0	-0.000
bone_marrow_samples	sampling_date	1034	100	0	0	-0.000
pcr	date	3102	0	0	100	79.000

table	kept-series mice with a row	% of kept-series mice
tissue_samples	449	86.500
blood_samples	443	85.400
bone_marrow_samples	448	86.300
cryo_samples	448	86.300
paraffin_samples	449	86.500
pcr	372	71.700

Columns with missing or empty values (27 of 75; the others are always filled in):

	table	column	% NULL	% empty	distinct
2	tissue_samples	tumour_id	1.600	0.000	807
4	tissue_samples	side_id	1.600	0.000	4
5	tissue_samples	tissue_type	0.100	0.000	18
7	tissue_samples	weight_g	1.700	0.000	217
8	tissue_samples	remark	0.000	8.000	214
9	tissue_samples	analysed	82.000	0.000	1
16	blood_samples	remark	0.000	95.900	13
17	blood_samples	analysed	40.600	0.000	1
18	blood_samples	type	78.200	0.000	1
25	bone_marrow_samples	remark	0.000	99.700	4
26	bone_marrow_samples	analysed	40.800	0.000	1
27	bone_marrow_samples	type	77.800	0.000	1
30	cryo_samples	tumour_id	3.200	0.000	799
33	cryo_samples	container	0.000	10.700	4
34	cryo_samples	tower	0.000	10.600	6
35	cryo_samples	drawer	0.100	99.200	4
36	cryo_samples	row	100.000	0.000	0
37	cryo_samples	column	100.000	0.000	0
42	cryo_samples	remark	0.000	24.200	171
48	paraffin_samples	cassette	51.000	0.800	228
49	paraffin_samples	container	100.000	0.000	0
50	paraffin_samples	drawer	100.000	0.000	0
54	paraffin_samples	remark	0.100	19.100	195
57	pcr	sample_id	0.000	7.400	1793
62	pcr	cell_count3	78.300	0.000	628
67	pcr	cp3	78.300	0.000	422
70	pcr	remark	0.000	78.000	10

What the measured lab values look like: tissue weights per site (`tissue_samples.weight_g`, `sites`) and the ALU-PCR results (`pcr`, human cells per sample, by sample type).

site	rows	mice	with_weight	weight_g
liver at necropsy	3249	802	3247	0.18 [0.10–0.30]
lung at necropsy	1601	803	1596	0.09 [0.06–0.11]
cerebrum at necropsy	1598	800	1596	0.20 [0.18–0.21]
primary tumour	796	734	796	0.33 [0.11–0.79]
spleen at necropsy	115	115	0	–
kidneys at necropsy	2	1	2	0.00 [0.00–0.00]
intestine at necropsy	1	1	0	–

sample_type	rows	mice	trails	cells	Cp	first	last
blood	626	619	0, 1, 3, 8, 10, 11, 12, 15	0.1247 [0.0049–13.5288]	23.1 [21.6–26.8]	2014-01-16	2021-08-22
bone marrow	618	613	0, 1, 3, 8, 10, 11, 12, 15	0.0208 [0.0007–1.4225]	25.6 [22.6–26.9]	2014-01-16	2021-08-24
brain	570	559	0, 3, 8, 10, 11, 12, 15	0.0016 [0.0000–0.3761]	25.5 [24.1–27.0]	2014-01-27	2021-08-25
liver	570	563	0, 3, 8, 10, 11, 12, 15	0.0128 [0.0004–0.6890]	25.7 [24.5–27.3]	2014-01-27	2021-08-25
lung	632	631	0, 3, 8, 10, 11, 12, 15	0.3962 [0.0026–19.6125]	22.1 [19.4–25.5]	2014-01-21	2021-08-23
spleen	118	112	11, 12, 15	0.0006 [0.0001–0.0025]	23.1 [22.7–24.0]	2021-09-13	2021-10-13

D3. The application log: an audit trail?

`app_log` is the log of the web application. Its `payload` column holds the request parameters (names and values translated). In the original export it **contains user names, session ids and passwords typed into the login form**; in the English copy passwords and session ids are masked. It is never displayed here; only counts and the parameters of the visit forms are extracted.

action	rows	first	last
show_login_site	72200	2020-05-08 06:21:56	2025-08-08 13:52:50
Querylog	52771	2020-05-08 06:28:40	2025-08-08 13:53:26
show_start_site_succ	48431	2020-05-08 06:29:47	2025-08-08 13:53:25
login_error	1101	2020-05-10 16:06:20	2025-08-08 13:52:53
login_succ	1063	2020-05-08 06:28:40	2025-08-08 13:52:56
logout_succ	716	2020-05-08 06:36:00	2025-06-23 05:30:36
show_start_site_error	637	2020-05-15 21:29:16	2024-12-02 17:53:17
1	2	2021-04-15 18:29:42	2021-04-15 18:29:45
fsh	1	2021-04-15 17:18:00	2021-04-15 17:18:00

```
app_log: 176,922 rows from 2020-05-08 to 2025-08-08; request rows with parameters
(Querylog): 52,771; visit form submissions with a weight: 23,013 new visits, 961 ed
its
```

Matching the logged visit forms with `visits`: a new-visit submission matches a stored visit of the same mouse at the same date and time (hh:mm); an edit names the stored visit (its visit id).

visit forms in the log	count
visits stored since the log starts	12168
… with a logged new-visit form (same mouse, date, hh:mm)	11564
… whose first logged weight differs from the stored weight	34
new-visit forms submitted more than once for the same mouse/date/time	8718
visits saved again through the edit form (distinct visit ids)	331
… still in visits	331
… stored weight equals the last logged edit	330
… first entered weight known and different from the stored one	16

trail	visits	median_abs_change_g	max_abs_change_g
11	1	1.000	1.000
12	23	2.000	200.000
15	7	0.600	2.300
16	3	0.900	1.300

trail	11	12	13	15	16	17	18
visits	192	10062	2	1286	501	9	116
with_logged_form	140	9567	0	1243	489	9	116

D4. Old-table columns v1 never used

Share of visits per trail in which a column has content (not NULL, not empty and not 0, so a score of 0 counts as "no content"). The single score-sheet items (`cond_*` : general condition, `beh_*` : behaviour) are summarised as "any item > 0".

% of visits with content:

trail	visits	rel. weight	weight points	any cond. item	cond. points	any beh. item	beh. points	tumour volume	tumour 2 volume	ulceration	tumour points	score	burden note	remark	time
0	35	97	34	0	0	0	0	66	0	0	66	66	0	0	100
1	1217	97	5	2	2	0	0	10	0	0	10	13	0	10	100
2	7	0	0	0	0	0	0	100	0	43	100	100	0	57	100
3	7047	95	5	1	1	0	0	43	0	0	43	44	0	26	100
6	13	0	0	0	0	0	0	92	0	0	92	92	0	0	100
7	39	92	0	8	8	0	0	77	0	8	85	85	0	23	100
8	428	100	1	0	0	0	0	54	53	0	54	58	0	13	100
9	11	0	0	0	0	0	0	100	0	0	100	100	0	0	100
10	7159	97	7	0	0	0	0	70	0	0	70	70	0	4	100
11	192	97	6	0	0	0	0	54	0	0	54	54	0	20	100
12	10062	98	1	2	2	0	0	30	0	0	30	32	0	15	100
13	2	50	0	0	0	0	0	100	0	0	100	100	0	0	100
15	1286	98	3	0	0	0	0	25	0	0	25	27	0	22	100
16	501	97	0	1	1	0	0	13	0	0	13	14	0	7	100
17	9	0	0	0	0	0	0	0	0	0	0	0	0	0	100
18	116	98	0	0	0	0	0	0	0	0	0	0	0	2	100

trail	primary_tumour_records	with_weight	weight_g
0	1	1	0.66 [0.66–0.66]
1	10	3	0.28 [0.21–0.49]
2	6	0	–
3	393	270	0.16 [0.05–0.50]
5	2	2	0.49 [0.34–0.65]
7	3	0	–
8	40	40	0.36 [0.24–0.77]
9	11	11	1.27 [0.89–1.45]
10	211	172	0.89 [0.59–1.14]
11	6	6	1.16 [0.69–1.32]
12	241	217	0.24 [0.12–0.47]
13	2	0	–
15	40	1	0.20 [0.20–0.20]
16	7	2	0.38 [0.36–0.39]

```
Daily device checks (irradiation device, not per mouse), rows per year: {2013: 57,
2014: 34, 2015: 7, 2016: 5, 2018: 18, 2019: 49, 2021: 37, 2022: 6, 2024: 9, 2025: 1
0}
```

What this tells us (Part D).

- Of the 30 new tables, 10 hold lab/sample data, 1 is the application log, 16 are lookups and 3 configure the web application.
- The lab tables are end-point data: tissue 99%, blood 100%, bone marrow 100% of the dated rows were taken within a day of death (necropsy); the `pcr` date is the analysis date, a median of 79 days after death. Only `tissue_samples.weight_g` (organ/tumour weight) and the ALU-PCR cell counts are measurements; the cryo, paraffin, blood and bone-marrow tables mostly record which samples exist and where they are stored. They could be per-mouse **targets** (metastasis burden) or static covariates, not time-varying inputs for weight forecasting.
- `app_log` is a **web request log, not a row-level audit trail**: no table, no old value, no database user per change. It starts on 2020-05-08, so it can only cover trails visited since then. Its form submissions still show corrections: 11,564 of 12,168 visits stored since then (95%) have a logged entry form, 34 of them with a first weight different from the stored one (up to 200 g), and 331 visits were saved again through the edit form. The original export stores plaintext passwords there; the English copy masks them.
- Beyond weight, the per-visit columns with content are the total score (non-zero in trail 1 13%, trail 3 44%, trail 10 70%, trail 12 32%, trail 15 27%) and the tumour volume (> 0 in trail 1 10%, trail 3 43%, trail 10 70%, trail 12 30%, trail 15 25%); the single score-sheet items are almost never set (at most 2% of the visits in these trails) and the burden note is empty. Score and tumour volume are the candidate time-varying covariates.

Part E: data quality and summary

E1. Data-quality issues, all trails, both exports

v1's list (A5), extended by duplicate visits, visits after death and treatments without a visit on the same day.

unit	issue	old export	new export	added
mice	**in trail 0, which is not in the trails table**	2	2	0
	with role 0 (not in the roles table)	8	8	0
	without a death date	63	34	-29
	with an injection date before birth	2	2	0
visits	**with an invalid date (e.g. 0000-00-00, day 00)**	5	5	0
	of a mouse not in the mice table	1	1	0
	before the birth of the mouse	29	29	0
	before the start of the trail	139	139	0
	more than 1 day after death	36	36	0
	with weight 0 g	10	10	0
	with weight > 100 g	11	11	0
	with weight between 0 and 10 g	1	1	0
	duplicates (same mouse, date and time)	142	188	46
	total score not the sum of the sub-scores	24	27	3
mouse-days	**with more than one visit**	304	352	48
treatments	**RT fractions without a visit that day**	27	27	0
	ChT administrations without a visit that day	23	23	0
	with an invalid date	0	0	0
trails	**without a start date**	1	1	0
	without mice	1	1	0

E2. Facts per export (as v1 Part D)

fact	old export (2023)	new export (2025)
Tables	25	55
Mice	843	988
Visits	22142	28125
Trails	15	18
Approved projects	12	15
Roles	11	14
Visits: first – last date	2013-06-17 – 2022-04-07	2013-06-17 – 2025-06-24
RT fractions / ChT administrations / surgeries	1958 / 1316 / 47	2158 / 1406 / 67
Mice with a death date	780 (93%)	954 (97%)
Trail 3, RT/RChT/ChT through michael.load_dataset: mice / rows	186 / 3,049	186 / 3,049
All trails, RT/RChT/ChT: mice / rows	434 / 12,397	519 / 16,376
All trails, RT/RChT/ChT: trails with ≥ 10 mice	1, 3, 10, 12	1, 3, 10, 12, 15

E3. Candidate datasets for modelling

Concrete options, all from the new export through `michael.load_dataset` (v1's filters; the last option with the treatment filter switched off). Forecast origins are counted as in the rolling evaluation (`MIN_CONTEXT` = 2: every row from the third one of a mouse on).

option	mice	rows	forecast origins	date span	trails	cell lines	% rows trail 3	% rows treated	naive MAE (g)
1. Trail 3, RT/RChT/ChT (v1 study cohort)	186	3049	2677	2013-08 – 2016-08	1	3	100	26	0.662
2. Trails 3 + 10 + 12, RT/RChT/ChT	453	13757	12851	2013-08 – 2025-02	3	8	22	18	0.341
3. All trails, RT/RChT/ChT	519	16376	15338	2013-06 – 2025-02	9	12	19	17	0.365
4. All trails, all roles, no treatment filter (context pool)	888	26974	25198	2013-06 – 2025-06	10	13	25	10	0.381

option	main caveat
1. Trail 3, RT/RChT/ChT (v1 study cohort)	one experiment, 186 mice; the 18 daily-weighed 2016 mice give 49% of the RT origins
2. Trails 3 + 10 + 12, RT/RChT/ChT	three experiments with different cell lines/strains (8 cell lines); trail 12 mixes two tumour models
3. All trails, RT/RChT/ChT	9 trails, many tiny pilots; protocols and populations differ, needs trail as a covariate
4. All trails, all roles, no treatment filter (context pool)	34% of the mice untreated (controls etc.); only as context, not as test series for treatment effects

E4. Open questions for the data owners

1. **Trail of a cohort vs trail of a mouse** (`cohorts.trail_id` vs `mice.trail_id`): for 229 of 987 mice with a batch the trail of their cohort differs from their own trail (e.g. the TAS102 cohorts point to trail 8). Which one is right, or does the cohort's trail mean something else?
2. **Trail 12 / PC-3DR**: the mice added to trail 12 come from cohort(s) PC-3DR RGB + luciferase main experiment (PC-3DR RGB + luciferase). Is this a second experiment under the same trail id, and should it be analysed separately?
3. **Missing death dates**: 34 mice have no death date (by trail: 1: 2, 2: 7, 4: 6, 10: 8, 13: 2, 17: 9). Are these mice still alive, transferred, or are the records incomplete? Michael's loader drops them.
4. **Trail 0 and role 0**: 2 mice in trail 0 and 8 with role 0. What are they?
5. **Edited records**: 38 death records of trail(s) 15 were completed after 2023 and 4 chemotherapy routes changed from "unknown" to i.p. Are further corrections to the old data expected, and is there a change history before 2020-05-08 (when the application log starts)?
6. **Weight corrections**: 34 visits since 2020-05-08 were stored with a weight different from the one first entered in the form. Which value is right, and were older weights corrected without a trace?
7. **Application log**: the original export holds plaintext passwords and session ids. Should it be removed from exports given to us?
8. **Invalid dates**: 5 visits have dates like `0000-00-00` or day `00`. Can they be repaired?
9. **Lab samples**: what do `analysed` ("yes") in the blood and bone-marrow samples and `special_sample` in `pcr` mean, and is the ALU-PCR cell count the intended metastasis read-out?
10. **Translation**: the English names were made for this notebook (glossary at the top). Some codes are uncertain, e.g. the tissue types "relk" and "leber_riecken" and the "_18" sample schemes; please confirm.

What this tells us (Part E).

- Most data-quality counts are the same in both exports (16 of 20 issues); the changes are without a death date mice: -29; duplicates (same mouse, date and time) visits: +46; total score not the sum of the sub-scores visits: +3; with more than one visit mouse-days: +48. The v1 problems "visits before the trail start" and "weights of 0 g / > 100 g" are unchanged, i.e. all in the old data.
- Candidate datasets range from 186 mice / 2,677 origins (trail 3) to 519 mice / 15,338 origins (all trails, RT/RChT/ChT) and 888 mice / 25,198 origins with untreated mice as context.
- Trails 3 + 10 + 12 hold 84% of the RT/RChT/ChT rows of all trails; the rest comes from trail 1 (1047 rows), trail 15 (900 rows), trail 16 (342 rows), trail 11 (181 rows), trail 8 (110 rows), trail 7 (39 rows).
- Naive last-value MAE differs by a factor of 4.0 between the kept trails (largest in trail 3, median gap 2 days; smallest in trail 10, median gap 1 day), so metrics on pooled data should also be reported per trail.